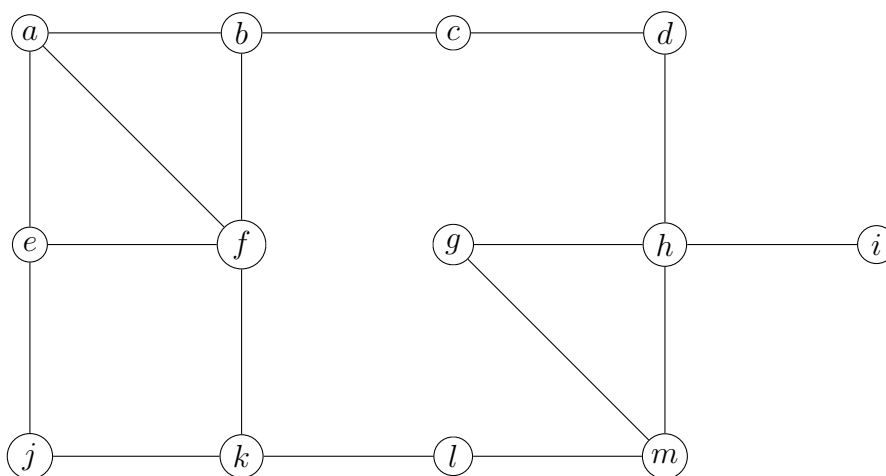


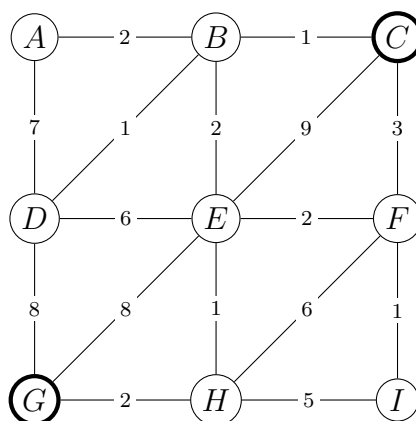
Group Names: _____

1. Consider the following graph.
 - (a) Which are the vertices of odd degree? _____
 - (b) **Eulerize this graph:** find the smallest number of edges you can duplicate so that you can construct an **Euler circuit**, and add them to the graph.
 - (c) Draw the circuit on the graph (offset slightly so you can see it) so that you can follow it without lifting your pencil from the circuit.



- 1

2. Consider the following weighted graph.



- (a) There are two vertices of odd degree in this graph, C and G . Use guess-and-check to find a path of shortest weighted distance between C and G . *Hint: the shortest-distance path has four edges.* (There is an algorithm that will do this, called Dijkstra's algorithm, which your textbook discusses — you can see why having a computer find shortest-distance paths might be useful!)

What is your shortest-weighted-distance path? _____

- (b) On the copy of the graph below, **duplicate your minimum distance path** (including the weights) to eulerize the graph. Then find an **Euler circuit** in the graph, which will be of minimum total weight.

